LAB 9: COMPONENT DIAGRAM - SYSTEM ARCHITECTURE

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AIM

To construct the UML Component Diagram of the Multi-Vendor E-Commerce Platform, showing physical software components, database modules, and external provider integration interfaces.

COMPONENTS & PROVIDED/REQUIRED INTERFACES

Component Name	Main Responsibility	Interfaces Provided / Used
Next.js Storefront App	Client UI for browsing, cart management, and customer portal.	Uses `IAuth`, `ICatalog`, `ICheckout`
Vendor Dashboard	Seller UI for listing products and processing sub-orders.	Uses `IVendorApi`, `IFulfilment`
Node.js API Gateway	Core backend service handling routing, RBAC, and business rules.	Provides REST Endpoints; Uses `IDataAccess`
Prisma ORM & Postgres DB	ACID relational storage for users, catalog, and settlements.	Provides `IDataAccess`
Payment SDK / Webhook	Third-party gateway handling payment cards & webhooks.	Provides `IPayment`

COMPONENT DIAGRAM

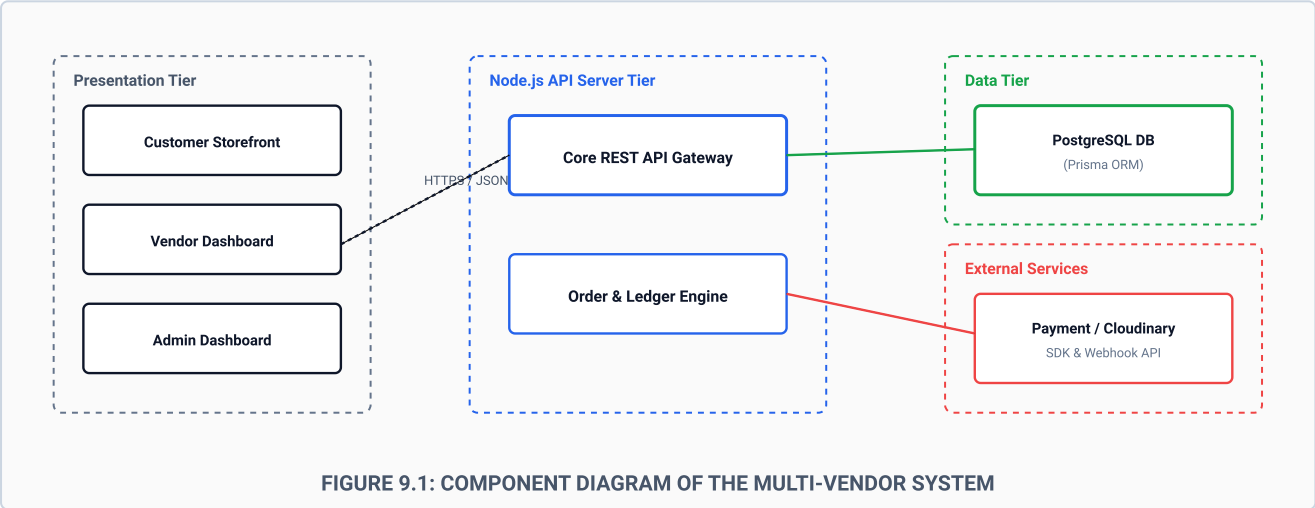


FIGURE 9.1: COMPONENT DIAGRAM OF THE MULTI-VENDOR SYSTEM

CONCLUSION

The UML Component Diagram details the software architecture for the Multi-Vendor E-Commerce Platform, establishing modular isolation between client portals, central API services, PostgreSQL storage, and payment integrations.